

# Notes: Chemmanur & Fulghieri (JF, 1994)

## Investment Bank Reputation, Information Production, and Financial Intermediation

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# 1 Big picture

- **Question.** How can investment banks credibly certify issuer quality when investors cannot observe the resources spent on evaluating firms?
- **Core friction.** Investment banks produce information, but their evaluation standard is unobservable. A bank can save evaluation costs by being lax, creating a moral-hazard problem in certification.
- **Core mechanism.** Reputation disciplines the intermediary. A bank that markets a bad firm today risks losing future credibility, future underwriting fees, and future surplus from certifying issuers.
- **Setting.** Entrepreneurs have projects and need outside equity. They can sell equity directly or through an investment bank. Investment banks repeatedly interact with the equity market, evaluate projects, and market equity for a fee.
- **Main equilibrium result.** Evaluation standards, market values of underwritten equity, underwriting fees, investment bank reputation, and entrepreneurs' choice between underwritten and direct equity sales all arise endogenously.
- **Key comparative static.** For low reputation values, a more reputable high-cost bank chooses a stricter standard. For very high reputation values, the relation reverses: a highly reputable bank may have weaker incentives to maintain strict standards because the reputational loss from a mistake is smaller.
- **Empirical implication.** More reputable investment banks reduce information asymmetry more effectively; IPO underpricing and negative seasoned-equity announcement effects should be smaller for more reputable underwriters.
- **Broader implication.** The same reputation logic applies to rating agencies, takeover advisors, convertible-call advisors, and other intermediaries that produce costly but unverifiable information.

# 2 Incremental contribution of the paper

- **Endogenizes certification credibility.** Earlier certification models often assume intermediaries can credibly signal quality. Chemmanur and Fulghieri model why credibility is possible when evaluation effort is unobservable and costly.
- **Links reputation to evaluation standards.** The paper does not treat reputation as an exogenous ranking. Reputation is a posterior belief about the intermediary's type, updated through the quality of firms previously marketed.
- **Models the intermediary's dynamic trade-off.** Investment banks trade off short-run evaluation costs against long-run reputational rents. This dynamic trade-off is the source of credible information production.
- **Generates underwriter choice endogenously.** Entrepreneurs decide whether to sell equity directly or through a bank based on the market value created by certification net of underwriting fees.
- **Produces non-monotone reputation incentives.** The paper shows that reputation can discipline banks at low and intermediate reputations but can create a "milking reputation" effect at very high reputations.
- **Connects theory to issuance evidence.** The model gives predictions for IPO underpricing, seasoned equity price effects, underwriting compensation, and the choice between underwritten and nonunderwritten offerings.
- **Generalizes beyond underwriting.** The paper shows how reputation can support credible intermediation in any market where expert verification is costly, noisy, and unverifiable to investors.

**Interpretation:** The incremental contribution is the reputation microfoundation. The paper explains not only that reputable investment banks certify issuers, but why investors believe their certification and why banks choose costly evaluation standards in equilibrium.

### 3 Model primitives and measurement

This is a theoretical paper. The relevant “measurement” objects are not sample variables but model primitives: investment bank type, evaluation standard, prior reputation, issuer quality, equity value, underwriting fees, and investor beliefs.

#### 3.1 Agents and timing

- **Entrepreneurs.** Each entrepreneur enters the equity market once and has either a good firm or a bad firm.
- **Investment banks.** Banks interact repeatedly with the equity market. They evaluate projects and report evaluations to investors.
- **Investors.** Investors set the market value of equity based on their beliefs about firm quality and investment bank credibility.
- **Dates.** The basic model has two dates. Time 0 actions can affect time 1 reputation and fees.

**Interpretation:** The repeated interaction is essential. Without future reputation at stake, costly evaluation would collapse because the high-cost bank would choose the cheapest possible standard.

#### 3.2 Firm quality and direct equity sale

Let  $f \in \{G, B\}$  denote firm type. A good firm has higher value than a bad firm. If the entrepreneur sells equity directly, investors observe only the prior probability of a good firm and value the equity accordingly. Denote the direct-sale value by  $u_t$ .

**Interpretation:** Direct sale pools good and bad firms. The investment bank is valuable because it can partially separate good firms from bad firms through evaluation.

#### 3.3 Investment bank types and reputation

There are two investment-bank types:

- a no-cost type  $N$ , which can evaluate firms at no cost;
- a high-cost type  $H$ , which bears a cost when setting a stricter evaluation standard.

The bank’s reputation at date 0 is investors’ prior probability that it is the no-cost type:

$$\alpha_0 = Pr(N).$$

**Interpretation:** Reputation is a belief, not an observable characteristic. It is updated based on the realized quality of firms the bank has marketed.

### 3.4 Evaluation technology

In the baseline one-sided-error model, an investment bank can mistakenly give a good evaluation to a bad firm. Let  $r$  denote the probability of incorrectly certifying a bad firm as good. The strictest available standard is  $p$ , so

$$r \in [p, 1], \quad p > 0.$$

A lower  $r$  means a stricter evaluation standard.

The no-cost bank sets the strictest standard:

$$r_N^* = p.$$

**Interpretation:** The high-cost bank's problem is to choose  $r_H$ . Lower  $r_H$  improves credibility but raises evaluation cost.

### 3.5 Investment bank fee

If the investment bank creates a surplus by selling equity at value  $V_t$  instead of the direct-sale value  $u_t$ , it receives a fraction  $k$  of this surplus:

$$Fee_t = k(V_t - u_t).$$

The entrepreneur's net proceeds from using the bank are

$$V_t - k(V_t - u_t).$$

**Interpretation:** Fees are endogenous to the value of certification. A more credible bank creates more surplus and can charge a larger fee.

## 4 Models and theoretical specifications

### 4.1 Investment bank objective

At time 1, the high-cost investment bank chooses its evaluation standard to maximize one-period profit:

$$\pi_H = k(V_1 - u_1) - C(r_H).$$

At time 0, the high-cost bank maximizes current fees plus expected future profit:

$$\Pi_H = k(V_0 - u_0) - C(r_H) + E_0[\pi_H].$$

**Interpretation:** The current cost of strict evaluation is paid immediately. The benefit arrives through reputation and future fees. This creates a dynamic incentive problem.

### 4.2 Entrepreneur's equity-marketing choice

The entrepreneur chooses the investment bank if net proceeds exceed direct-sale proceeds:

$$V_t - k(V_t - u_t) > u_t.$$

Otherwise, the entrepreneur sells equity directly.

**Interpretation:** The bank is used only when certification increases value enough to cover the fee. This is why the choice between underwritten and nonunderwritten equity sales is endogenous.

### 4.3 Sequential equilibrium

An equilibrium consists of:

- the investment bank's evaluation standard;
- the entrepreneur's decision to use a bank or sell directly;
- investors' beliefs about the bank's type and the firm's type;
- equity values and underwriting fees.

Beliefs are updated by Bayes' rule on the equilibrium path.

**Interpretation:** Investors use the bank's past outcomes to infer whether it is likely to be a no-cost or high-cost evaluator. This is the formal reputation mechanism.

### 4.4 Proposition 1: equilibrium structure

**Proposition 1: Equilibrium strategy-belief structure**

The no-cost investment bank sets the strictest standard  $p$ . The high-cost investment bank sets a stricter standard at time 0 than it would in a one-period game because time 0 mistakes damage reputation and future profits. Entrepreneurs use the bank after good evaluations when certification raises net proceeds above direct-sale proceeds.

**Interpretation:** The proposition establishes that investment banks can act as credible information producers even though their evaluation standards are unobservable. Reputation induces costly evaluation.

### 4.5 Proposition 2: interior high-cost standard

**Proposition 2: Interior evaluation standard**

Under a parameter restriction on the marginal cost of stricter evaluation, the high-cost investment bank chooses a unique interior standard  $r_H^*$  satisfying  $1 > r_H^* > p$ .

**Interpretation:** The high-cost bank is neither fully lax nor fully strict. The interior standard balances the marginal cost of investigation against the marginal benefit of protecting reputation.

### 4.6 Proposition 3: effect of reputation

**Proposition 3: Non-monotone reputation effect**

For low reputation values, a higher reputation makes the high-cost investment bank choose a stricter evaluation standard. For reputation values close to one, a higher reputation makes the high-cost bank choose a less strict standard.

**Interpretation:** Reputation has a non-monotone discipline effect. When the market is uncertain about the bank's type, a mistake is very informative and costly. When reputation is already near perfect, a mistake is less damaging, so the bank can be tempted to milk its reputation.

## 4.7 Proposition 4: comparative statics

### **Proposition 4: Fee share, evaluation cost, and variance**

The high-cost bank chooses a stricter standard when the underwriting fee share  $k$  rises and a less strict standard when the marginal cost  $c$  of strict evaluation rises. For low reputation values and sufficiently high good-firm probability, the variance of firms marketed by the high-cost bank declines as reputation rises.

**Interpretation:** Higher expected rents from reputation encourage stricter evaluation. Higher evaluation costs discourage it. This result links underwriting compensation to certification quality.

## 4.8 Generalized setting

The paper extends the basic model in two directions:

1. investment banks can make two-sided errors, rejecting good firms as well as certifying bad firms;
2. there are multiple banks with different reputation levels, and rejected entrepreneurs can approach the next-best bank.

In equilibrium, entrepreneurs first approach the bank with the highest reputation. If rejected, they move down the reputation ladder. Direct equity sale occurs only after no bank markets the issue.

**Interpretation:** The generalized model makes the underwriting hierarchy explicit. It predicts that more reputable banks market better pools of firms and that equity value increases with underwriter reputation.

## 5 Theoretical design and identification logic

This is a theory paper, so the relevant “identification” is the model design that isolates the credibility problem in information production.

### 5.1 What the model isolates

The model isolates three frictions:

- **asymmetric information** between entrepreneurs and outside investors;
- **moral hazard** by investment banks because evaluation standards are unobservable;
- **dynamic reputation** because banks repeatedly interact with investors.

**Interpretation:** Removing any one of these elements changes the mechanism. Without asymmetric information, certification is unnecessary. Without moral hazard, certification is automatically credible. Without repetition, costly evaluation has no future reward.

### 5.2 Benchmark comparison

The key comparison is with a one-period game. In a one-period game, the high-cost bank minimizes evaluation cost and sets  $r_H = 1$ . Information production has little credibility. In the dynamic game, the bank sets  $r_H < 1$  when reputation matters.

$$\text{one-period game: } r_H = 1, \quad \text{dynamic reputation game: } r_H^* < 1.$$

**Interpretation:** This comparison identifies the role of reputation. The same high-cost bank behaves differently when future reputation and fees are at stake.

### 5.3 Why reputation matters for security prices

Investors value equity using beliefs about both firm quality and bank credibility. A good evaluation from a more reputable bank is more informative, so investors pay more for the equity. This increases net proceeds to entrepreneurs and fees to investment banks.

**Interpretation:** This gives a pricing interpretation of reputation. Reputation is not simply prestige; it changes the posterior probability that the issuer is good and therefore changes the market value of securities.

### 5.4 Empirical implications

The model predicts:

- IPO underpricing should be lower for offerings underwritten by more reputable investment banks;
- negative announcement effects for seasoned equity issues should be smaller for more reputable underwriters;
- underwriting fees and issuer proceeds should be increasing in underwriter reputation when reputation is in the disciplining region;
- direct or rights offerings should be used mainly by firms facing little information asymmetry or by firms unable to attract a reputable intermediary.

**Interpretation:** These predictions connect the formal model to the empirical literature on underwriter reputation, IPO pricing, SEO announcement effects, and the underwriting/direct-sale choice.

### 5.5 Limits of the theory

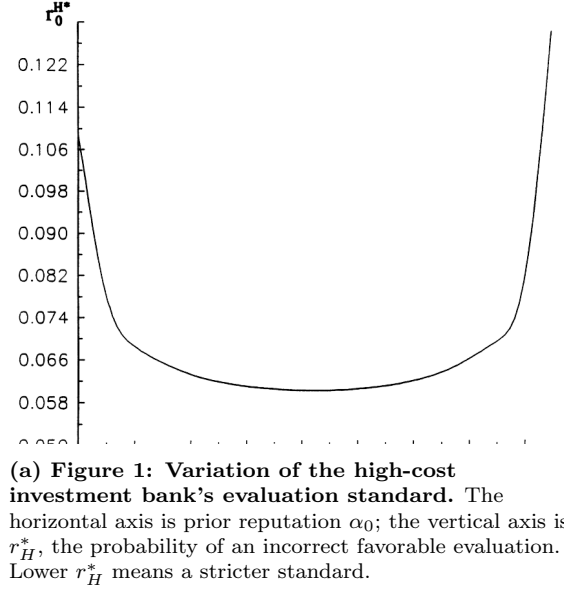
- The baseline has two dates and two bank types; richer dynamics could alter exact comparative statics.
- Evaluation technology is stylized, especially in the one-sided-error baseline.
- The model abstracts from underpricing mechanics, book-building, allocation rules, and legal liability.
- Reputation is summarized by a single posterior probability, while real underwriter reputation may have multiple dimensions.

**Interpretation:** The paper's value is conceptual. It provides a reputation-based foundation for credible certification, not a complete institutional model of underwriting practice.



## 6 Figures and model-result interpretation

### 6.1 Figure 1: Variation of the high-cost investment bank's evaluation standard with prior reputation



| Result            | Reading rule                                                                                   |
|-------------------|------------------------------------------------------------------------------------------------|
| Low reputation    | Higher reputation makes mistakes more costly; the high-cost bank sets a stricter standard.     |
| Middle reputation | Reputation is most informative; discipline is strongest.                                       |
| High reputation   | The bank may milk reputation; a mistake is less damaging, so the standard becomes less strict. |
| Fee share $k$     | Higher rents from reputation imply stricter evaluation.                                        |
| Cost $c$          | Higher evaluation cost implies laxer evaluation.                                               |

(b) **Notes-created interpretation table.** This table summarizes the model's comparative statics displayed in Figure 1 and in Propositions 3 and 4.

**Figure 1 and proposition summary.** The figure is directly cropped from the original paper. The paired table is a notes-created reading guide.

#### Read:

- The U-shaped curve plots  $r_H^*$  against the prior reputation  $\alpha_0$ .
- Since  $r_H^*$  is the probability of incorrectly evaluating a bad firm as good, lower values mean stricter evaluation.
- For low reputation, the curve slopes down: reputation strengthens incentives.
- For high reputation, the curve slopes up: the bank can exploit its reputation because an occasional mistake is less informative.

**Economic message:** The figure is the paper's central visual result. Reputation is not mechanically monotone. It can discipline information production, but very high reputation can also create a temptation to economize on evaluation effort.

**Interpretation:** The non-monotonicity is important because it challenges the simple intuition that the most prestigious investment banks always set the strictest standards. In the model, strict standards are most attractive when a mistake would meaningfully change investors' beliefs about the bank's type. If reputation is already near perfect, the market may forgive or discount a mistake.

## 6.2 Notes-created Table: Main model results and economic implications

| Result            | Formal mechanism                                                                                       | Economic implication                                                         |
|-------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Proposition 1     | Sequential equilibrium with reputation-based beliefs and endogenous evaluation standards.              | Costly evaluation is credible because future reputation matters.             |
| Proposition 2     | High-cost bank chooses unique interior $r_H^*$ under a cost restriction.                               | The bank balances current evaluation cost against future reputational rents. |
| Proposition 3     | $r_H^*$ falls with reputation at low $\alpha_0$ but rises near $\alpha_0 = 1$ .                        | Reputation disciplines at first, but very high prestige can be milked.       |
| Proposition 4     | $r_H^*$ decreases with fee share $k$ and increases with evaluation cost $c$ .                          | Compensation design and evaluation cost affect certification quality.        |
| Generalized model | Entrepreneurs approach more reputable banks first and move down the reputation ladder after rejection. | Underwriter reputation ranks the market and affects equity values.           |

**Notes-created table: Main model results and economic implications.** The original paper has no empirical tables. This table summarizes the theoretical propositions in the compact note format.

## 7 Short summary

- The paper models investment banks as repeat players that produce costly, noisy information about issuers.
- Investors cannot observe evaluation standards, so there is moral hazard in certification.
- Reputation is investors' posterior belief that a bank is a low/no-cost high-standard evaluator.
- A bad firm marketed by the bank damages reputation and future fees.
- Dynamic reputation induces high-cost banks to set stricter standards than they would in a one-period game.
- Market values of underwritten equity, bank fees, issuer proceeds, and direct-versus-underwritten choice are endogenous.
- For low reputation levels, a higher reputation induces stricter evaluation; near perfect reputation, the relation reverses.
- Higher underwriting rents induce stricter evaluation, while higher evaluation costs induce laxer evaluation.
- The generalized model predicts that entrepreneurs first approach the most reputable bank and then move down the reputation hierarchy.
- The model provides a foundation for empirical links between underwriter reputation, IPO underpricing, SEO announcement effects, and security valuation.

## 8 Conclusion

### 8.1 Core conclusion

Chemmanur and Fulghieri show how reputation can make financial intermediation credible when information production is costly and unverifiable. Investment banks certify issuers not because they are automatically trustworthy, but because poor certification damages future reputation and future fee income.

## 8.2 Economic intuition

A bank faces a dynamic trade-off. By spending more on evaluation today, it reduces the risk of marketing a bad firm as good. Avoiding such mistakes protects reputation and future rents. However, strict evaluation is costly. The equilibrium evaluation standard balances these forces. The market price of an underwritten issue reflects investors' beliefs about this standard.

## 8.3 Incremental contribution revisited

The paper's central contribution is to endogenize intermediary credibility. It derives reputation, evaluation standards, equity values, fees, and underwriting choice from a single dynamic game. This turns underwriter reputation from an empirical control variable into an economic mechanism.

## 8.4 Implications for underwriting markets

The model predicts that reputable banks certify better, obtain higher equity values for issuers, and earn fees from the surplus they create. It also predicts that firms facing more information asymmetry should prefer underwritten issues, while firms with little asymmetric information can use direct or rights offerings.

## 8.5 Limitations and extensions

The model is stylized. It abstracts from book-building, legal liability, roadshows, allocation discretion, analyst coverage, and complex underwriting contracts. It also treats reputation as a one-dimensional posterior belief. These simplifications make the reputation mechanism clear, but empirical applications must account for richer institutional features.

## 8.6 Takeaway

The enduring message is that reputation is a productive asset in financial intermediation. It allows investment banks to sell credible information when evaluation effort cannot be observed. But the same reputation can also be exploited, so reputation disciplines intermediaries most strongly when investors still have something meaningful to learn about their type.